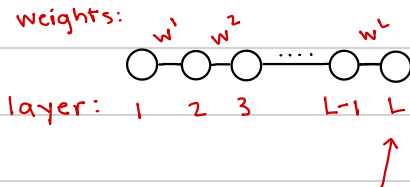


①

Backpropagation

Simple network with equations



$$z^L = w^L a^{L-1} + b^L$$

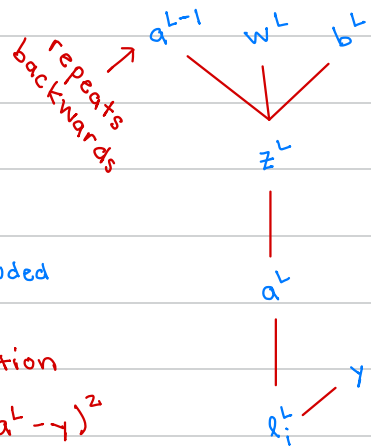
$a^L = \sigma(z^L)$ if included

σ : activation function

cost/error: $l_i^L = (a^L - y)^2$

MSE

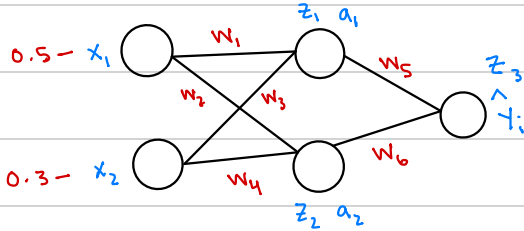
visualization



"Learning" is finding the right weights
and biases for the problem

②

Backpropagation steps, step #1



$$\sigma(z) = \text{sigmoid}$$

$$l = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

$$\text{learning rate: } \eta = 0.1$$

Weights are set randomly:

$$w_1 = 0.7$$

$$w_4 = 0.6$$

$$w_2 = 0.3$$

$$w_5 = 0.55$$

$$w_3 = 0.4$$

$$w_6 = 0.45$$

3 steps:

1. Feed forward

2. Loss calculation

3. Backpropagation

#1 $z_1 = (0.5 \cdot 0.7) + (0.3 \cdot 0.4)$

$$a_1 = \sigma(z_1) = 0.615$$

$$z_2 = (0.5 \cdot 0.3) + (0.3 \cdot 0.6)$$

$$a_2 = \sigma(z_2) = 0.582$$

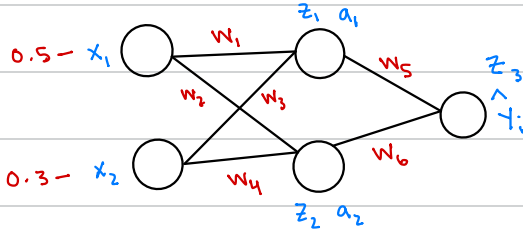
$$z_3 = (0.615 \cdot 0.55) + (0.582 \cdot 0.45)$$

$$\hat{y}_1 = \sigma(z_3) = \boxed{0.645}$$

Feed forward

②

Step #2



$$\sigma(z) = \text{sigmoid}$$

$$l = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

$$\text{learning rate: } \eta = 0.1$$

Weights are set randomly:

$$w_1 = 0.7$$

$$w_4 = 0.6$$

$$w_2 = 0.3$$

$$w_5 = 0.55$$

$$w_3 = 0.4$$

$$w_6 = 0.45$$

3 steps:

1. Feed forward

2. Loss calculation

3. Backpropagation

#2 only one output neuron

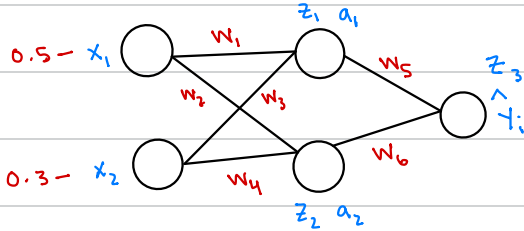
$$l = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2 \quad y \text{ (real label)} = 1$$

$$l = (1 - 0.645)^2 = \boxed{0.126}$$

Calculate the loss

3

Chain Rule, update w_5



$\sigma(z) = \text{sigmoid}$

$$l = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

learning rate: $\eta = 0.1$

Weights are set randomly:

$$w_1 = 0.7$$

$$w_4 = 0.6$$

$$w_2 = 0.3$$

$$w_5 = 0.55$$

$$w_3 = 0.4$$

$$w_6 = 0.45$$

3 steps:

1. Feed forward

2. Loss calculation

3. Backpropagation

#3 Backpropagation

start by updating w_5 using gradient descent

$$\text{new } w_5 = \text{current } w_5 - \eta \frac{\partial l}{\partial w_5}$$

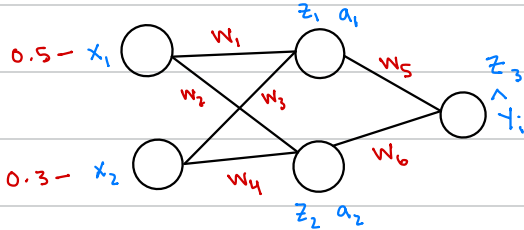
$$\text{chain rule: } \frac{\partial l}{\partial w_5} = \frac{\partial l}{\partial \hat{y}_1} \cdot \frac{\partial \hat{y}_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_5}$$

$$\textcircled{1} \quad \frac{\partial l}{\partial \hat{y}_1} \quad l = (y - \hat{y}_i)^2 \quad \frac{\partial l}{\partial \hat{y}_1} = -2(y - \hat{y}_i) = -0.71$$

next page

3

Chain Rule



$$\sigma(z) = \text{sigmoid}$$

$$l = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

$$\text{learning rate: } \eta = 0.1$$

Weights are set randomly:

$$w_1 = 0.7$$

$$w_4 = 0.6$$

$$w_2 = 0.3$$

$$w_5 = 0.55$$

$$w_3 = 0.4$$

$$w_6 = 0.45$$

3 steps:

1. Feed forward

2. Loss calculation

3. Backpropagation

#3 Backpropagation

start by updating w_5 using gradient descent

$$\text{new } w_5 = \text{current } w_5 - \eta \frac{\partial l}{\partial w_5}$$

$$\text{chain rule: } \frac{\partial l}{\partial w_5} = \frac{\partial l}{\partial \hat{y}_1} \cdot \frac{\partial \hat{y}_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_5}$$

$$\textcircled{2} \frac{\partial \hat{y}_1}{\partial z_3}$$

$$\hat{y}_1 = \sigma(z_3)$$

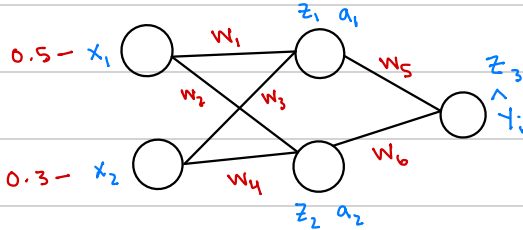
$$z_3 = 0.645$$

derivative of sigmoid $\sigma(z_3)(1 - \sigma(z_3))$

$$= 0.229$$

4

Chain Rule



$$\sigma(z) = \text{sigmoid}$$

$$l = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

$$\text{learning rate: } \eta = 0.1$$

Weights are set randomly:

$$w_1 = 0.7$$

$$w_4 = 0.6$$

$$w_2 = 0.3$$

$$w_5 = 0.55$$

$$w_3 = 0.4$$

$$w_6 = 0.45$$

0.551

3 steps:

1. Feed forward

2. Loss calculation

3. Backpropagation

#3 Backpropagation

start by updating w_5 using gradient descent

$$\text{new } w_5 = \text{current } w_5 - \eta \frac{\partial l}{\partial w_5}$$

$$\text{chain rule: } \frac{\partial l}{\partial w_5} = \frac{\partial l}{\partial \hat{y}_1} \cdot \frac{\partial \hat{y}_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_5}$$

$$\textcircled{3} \quad \frac{\partial z_3}{\partial w_5} \quad z_3 = a_1 \cdot w_5 + a_2 \cdot w_6$$

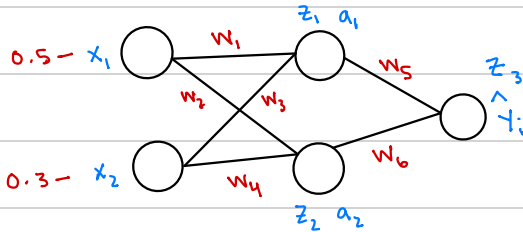
$$\frac{\partial z_3}{\partial w_5} = a_1 = 0.615$$

$$\frac{\partial l}{\partial w_5} = -0.71 \cdot 0.229 \cdot 0.615 = -0.01 \quad \text{update } w_5:$$

$$\text{new } w_5 = 0.55 + 0.01 \cdot 0.1 = 0.551$$

5

Backpropagation: update all $w_1 - w_6$



$\sigma(z) = \text{sigmoid}$

$$L = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

learning rate: $\eta = 0.1$

Weights are set randomly:

$$\begin{aligned} w_1 &= 0.7 & w_4 &= 0.6 \\ w_2 &= 0.3 & w_5 &= 0.55 \\ w_3 &= 0.4 & w_6 &= 0.45 \end{aligned}$$

update w_1 using the chain rule

$$\frac{\partial L}{\partial w_1} = \underbrace{\frac{\partial L}{\partial \hat{y}}}_{-0.71} \cdot \underbrace{\frac{\partial \hat{y}}{\partial z_3}}_{0.229} \cdot \frac{\partial z_3}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

① ② ③

$$\textcircled{1} \quad z_3 = w_5 \cdot a_1 + w_6 \cdot a_2 \rightarrow \frac{\partial z_3}{\partial a_1} = w_5 = 0.55$$

$$\textcircled{2} \quad a_1 = \sigma(z_1) \rightarrow \sigma'(z_3) = \sigma(z_3)(1 - \sigma(z_3)) = 0.615(1 - 0.615) = 0.237$$

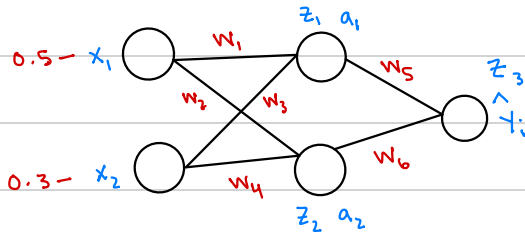
$$\textcircled{3} \quad z_1 = w_1 \cdot x_1 + w_3 \cdot x_2 \rightarrow \frac{\partial z_1}{\partial w_1} = x_1 = 0.5$$

$$\frac{\partial L}{\partial w_1} = -0.71 \cdot 0.229 \cdot 0.55 \cdot 0.237 \cdot 0.55 = -0.0106$$

$$\text{update } w_1: w_1 = 0.7 + (0.0106 \cdot 0.1) = 0.7010$$

6

After updating all weights



New weights:

$$w_1 = 0.701$$

$$w_4 = 0.6005$$

$$w_2 = 0.3009$$

$$w_5 = 0.551$$

$$w_3 = 0.4011$$

$$w_6 = 0.4595$$

Is the estimate $\hat{y}$ better with these weights?

$$x_1 = 0.5, x_2 = 0.3, y = 1 \rightarrow \hat{y} = 0.6443$$

$$l = (1 - \hat{y})^2 = 0.1245$$

Previous $l = 0.126$

YES!!